



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

INDEX
NUMBER

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BIOLOGY

Paper 2 Core Paper

9648/02

FRIDAY 26 AUGUST 2011

2 Hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A]

Answer all questions.

SECTION B]

Answer any **one** question.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	80
1 [11]	
2 [10]	
3 [12]	
4 [14]	
5 [11]	
6 [10]	
7 [12]	
Section B	20
8a / 9a [6]	
8b / 9b [7]	
8c / 9c [7]	
TOTAL	100

Section A

Answer **all** the questions in this section.

- 1 Microtubules are an important component of the cytoskeleton. They are formed by the assembly of two types of protein subunits known as α -tubulin and β -tubulin into protofilaments. These protofilaments are then bundled into hollow cylindrical filaments.

Fig. 1.1 illustrates the formation and breakdown of the protofilaments and the structure of the mature microtubule filament.

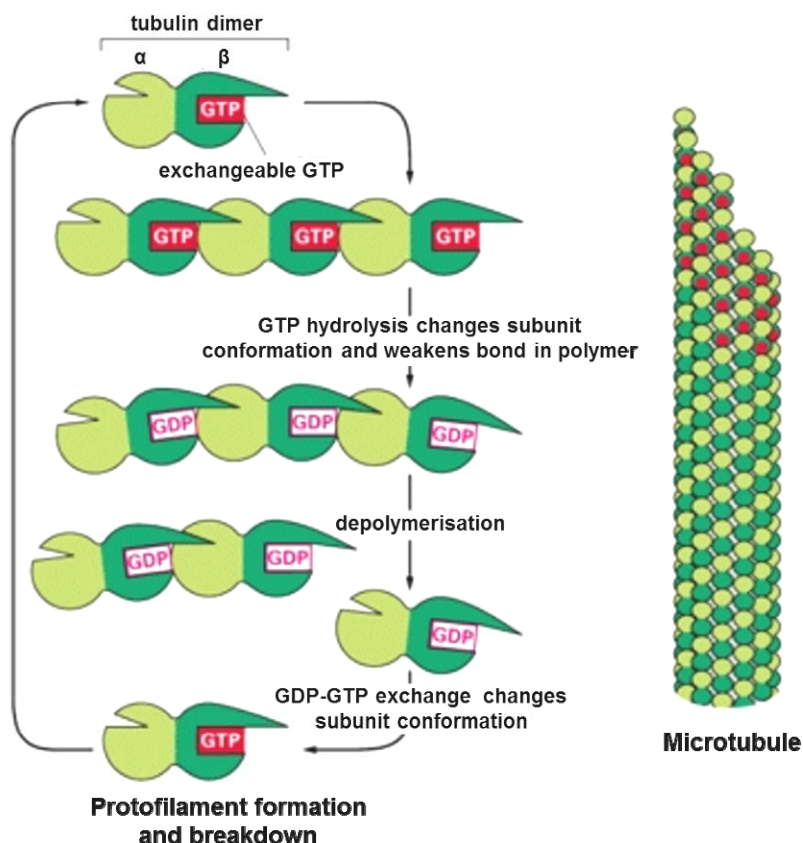


Fig. 1.1 (adapted from Molecular Biology of the Cell, 3rd edition)

(a) With reference to Fig. 1.1, describe **two** ways in which microtubules and cellulose

(i) are similar;

- 1:
- 2: [2]

(ii) are different.

- 1:
- 2: [2]

Microtubules are important in eukaryotic cell division processes including meiosis. Fig. 1.2 illustrates the behaviour of various structures during meiosis in the absence (control) and presence of taxane, a meiotic inhibitor. The curves show changes in distance between:

- A the centromere and the poles of the spindle
- B the centromeres of homologous chromosomes
- C the poles of the spindle

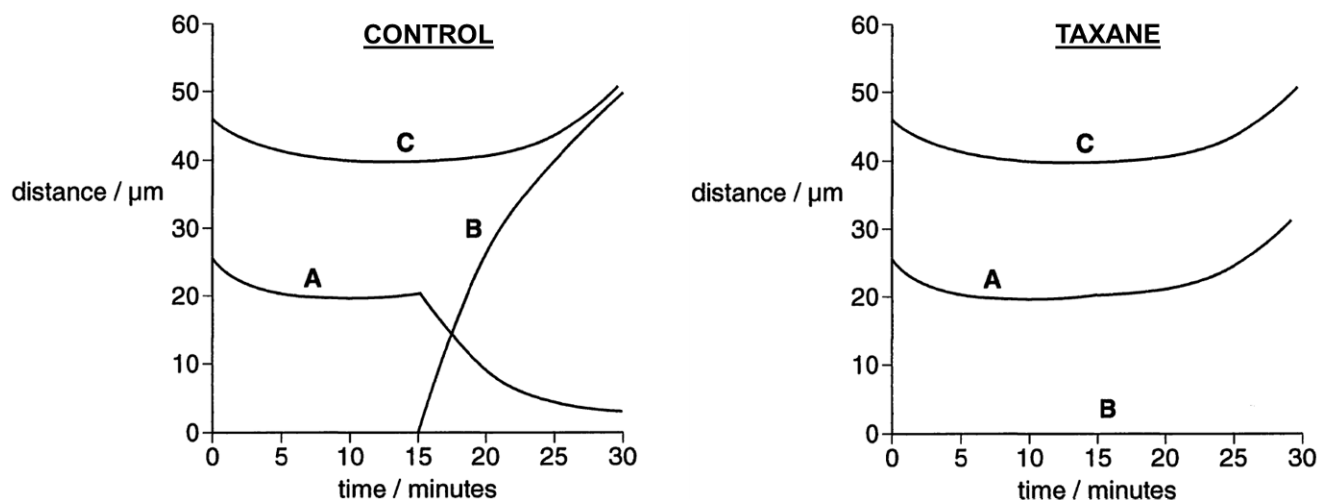


Fig. 1.2

(b) Describe and explain **two** reasons why meiosis is important in sexual reproduction.

- 1:
-
-
- 2:
-
-[4]

(c) With reference to Fig. 1.2, explain how taxane acts as a meiotic inhibitor.

-
-
-
-
-[3]

[Total: 11]

-
- Diagram illustrating the structure and function of RNA polymerase during transcription. The enzyme is shown as a large blue structure with a central channel. The DNA double helix (orange) enters from the top left, passes through the enzyme, and exits at the top right. The RNA transcript (blue) is being synthesized in the active site and exits through the RNA exit channel. Labels include: 3' 5', RNA polymerase, DNA double helix, DNA rewinding, jaws in close configuration, A, direction of transcription, flap in closed position, RNA exit channel, active site, newly synthesised RNA transcript, and short region of DNA-RNA helix.

With reference to Fig. 2.2,

- (i) Suggest why RNA polymerase begins transcribing the gene even before the first translated codon.

.....
.....[1]

- (ii) Describe what would happen to the resulting polypeptide chain if there was a mutation in the DNA sequence such that the base indicated in the **oval** was deleted.

.....
.....
.....[2]

- (c) The researchers managed to isolate two different mutant strains of *B.atlanticus*: the “quick-stop” strain and the “slow-stop” strain. The “quick-stop” strain is unable to carry out DNA synthesis whereas the “slow-stop” strain ceases DNA synthesis *after one round* of replication.

It was hypothesised that these results are due to defects in one of the following enzymes that are involved in DNA replication:

- Helicase
- RNA primase
- DNA ligase

Suggest and explain, in detail, which of the above enzymes is defective in the “**slow-stop**” strain.

.....
.....
.....
.....
.....[3]

[Total: 10]

3 (a) With reference to the *lac* operon, explain what is meant by :

(i) regulatory genes;

.....
.....
.....[2]

(ii) structural genes.

.....
.....
.....[2]

(b) Fig. 3 below is an electron micrograph of a microorganism that is isolated from the stools of cholera patients.



Fig. 3

With reference to Fig. 3, state **two** structural differences between the chromosome of this type of microorganism and a human cell.

1:
.....
2:
.....[2]

- (c) The enzyme lactase is responsible for lactose metabolism in humans and is encoded by a gene on chromosome 2.

Based on your understanding of eukaryotic and prokaryotic genomes, describe how the genes in humans and *Escherichia coli* that are responsible for lactose metabolism differ in terms of:

- (i) their organisation;

.....
.....[1]

- (ii) the control of gene expression.

.....
.....
.....
.....[2]

- (d) Streptomycin is an antibiotic that binds to bacterial ribosomes.

- (i) Explain how this effect of streptomycin prevents the growth of bacteria.

.....
.....
.....[2]

- (ii) Suggest why streptomycin does not affect the growth of human cells.

.....
.....[1]

[Total: 12]

4 (a) Define cancer.

.....

.....

.....

.....

.....[3]

Chronic myelogenous leukemia (CML) is a form of leukemia which arises from mutations to a bone marrow stem cell. Leukemic cells in many CML cases contain chromosome aberrations in chromosome 9 and chromosome 22 where a reciprocal translocation occurred. This results in the formation of a longer chromosome 9 and a shorter chromosome 22 also known as the Philadelphia (Ph) chromosome.

Most of DNA from the gene c-ABL originally found in chromosome 9 is fused with part of a gene BCR found in chromosome 22. This results in the formation of a BCR-ABL hybrid gene and leads to the translation of an abnormal fusion protein that constitutively activates a series of cellular processes which are normally only stimulated in the presence of growth factors. Fig. 4 illustrates the formation of the Ph chromosome.

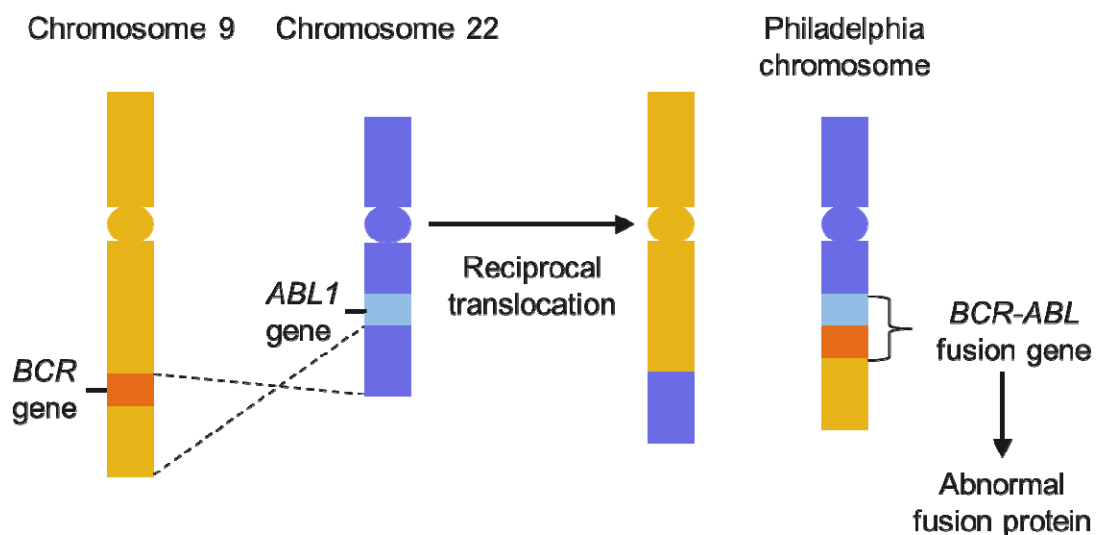


Fig. 4

(b) Identify and describe the function of the category of genes that c-ABL and BCR belong to.

.....

.....

.....[2]

- (c) Apart from the genetic mutation shown above, describe and explain **one** other type of genetic change in genes similar to c-Abl and BCR that can also result in abnormal stimulation of the cell cycle.

.....

.....

.....

.....

.....[3]

Cells containing the Ph chromosome have an increased rate of mitosis, however, more mutations are required before the patient's condition aggravates into a chronic stage.

- (d) Describe and explain how mutations to a **named** tumour suppressor gene can result in a significant reduction in rate of apoptosis in these cells.

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.....

.....[4]

The ABL protein and BCR-ABL fusion protein are tyrosine kinases, i.e. enzymes which phosphorylate and activate other proteins. Treatment with a new drug imatinib mesylate seems to cure the disease through blocking the binding of ATP to the active site of BCR-ABL fusion protein.

- (e) Suggest how imatinib mesylate reduces the activity of BCR-ABL.

.....

.....

.....[2]

[Total: 14]

- 5 Blood type in humans is determined by glycoproteins found on the surface of red blood cells. These glycoproteins are known as antigens. The ABO system of blood typing assigns individuals to blood types A, B, AB and O based on the presence of the A and B antigens.

Fig. 5.1 below shows the four ABO blood groups in humans and their corresponding antigens.

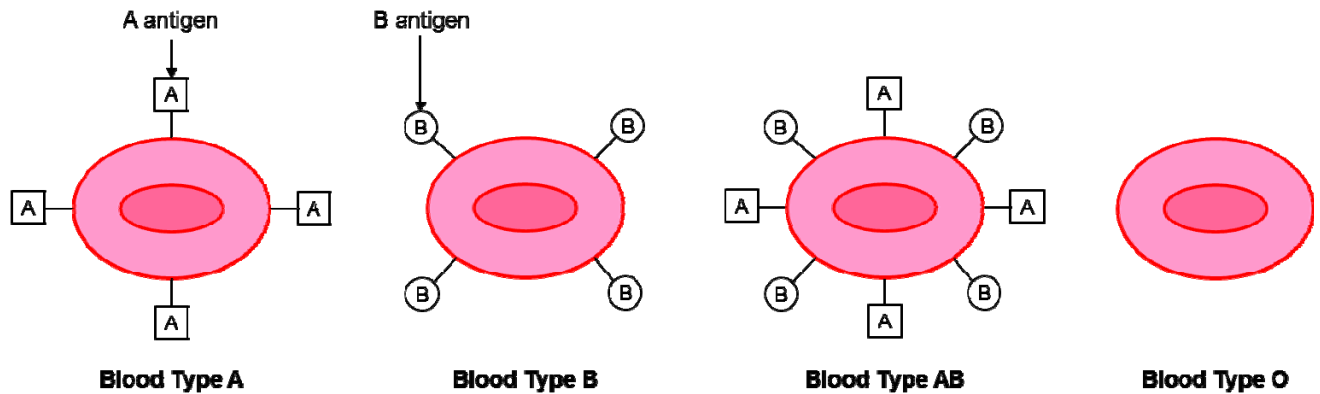


Fig. 5.1

The Bombay phenotype is a recessive condition causing affected individuals to appear as blood type O. The dominant allele H codes for an enzyme that produces the H antigen on the surface of red blood cells. The recessive allele h codes for a non-functional enzyme.

The H antigen is the precursor of both the A and B antigens. It is modified by enzymes coded by the ABO locus as follows:

I^A allele: A antigen I^B allele: B antigen I^O allele: H antigen (no modification)

- (a) With reference to the information provided, describe what is meant by discontinuous variation.

.....

[2]

Fig. 5.2 is a pedigree showing the blood types in a family of five individuals.

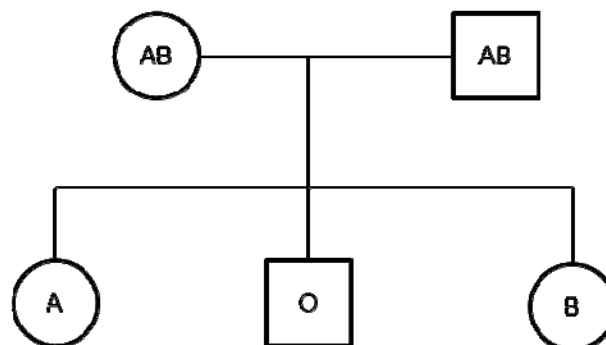


Fig. 5.2

- (b) (i)** Draw a genetic diagram in the space below to show how the different blood types in the family were obtained.

[4]

- (ii)** Explain the mode of inheritance shown in the pedigree.

.....
.....
.....[2]

- (c)** Calculate the probability that the couple's fourth child will be a girl designated with blood type O. Show your working clearly.

[1]

- (d)** A scientist researching the Bombay phenotype discovered a similar condition in mice known as "Mice-bay". The scientist suspected that the condition is sex-linked. She created a pure-breeding mutant strain of mice with the "Mice-bay" phenotype. These mice were used in crossing experiments with other strains of mice.

Suggest what kind of crossing experiments may be carried out to determine if the condition is sex-linked. Explain your answer.

.....
.....
.....[2]

[Total: 11]

- 6 Fig. 6.1 shows a computer simulation of synaptic transmission. The top left panel is a diagram of the cells in the synapse. The bottom left panel shows controls for firing (stimulation) by the presynaptic cells. The right panel is a graph showing the change in membrane potential (E_m) at the axon hillock of cell C. Figs. 6.2 and 6.3 different settings for the presynaptic cells and the corresponding E_m changes in cell C.

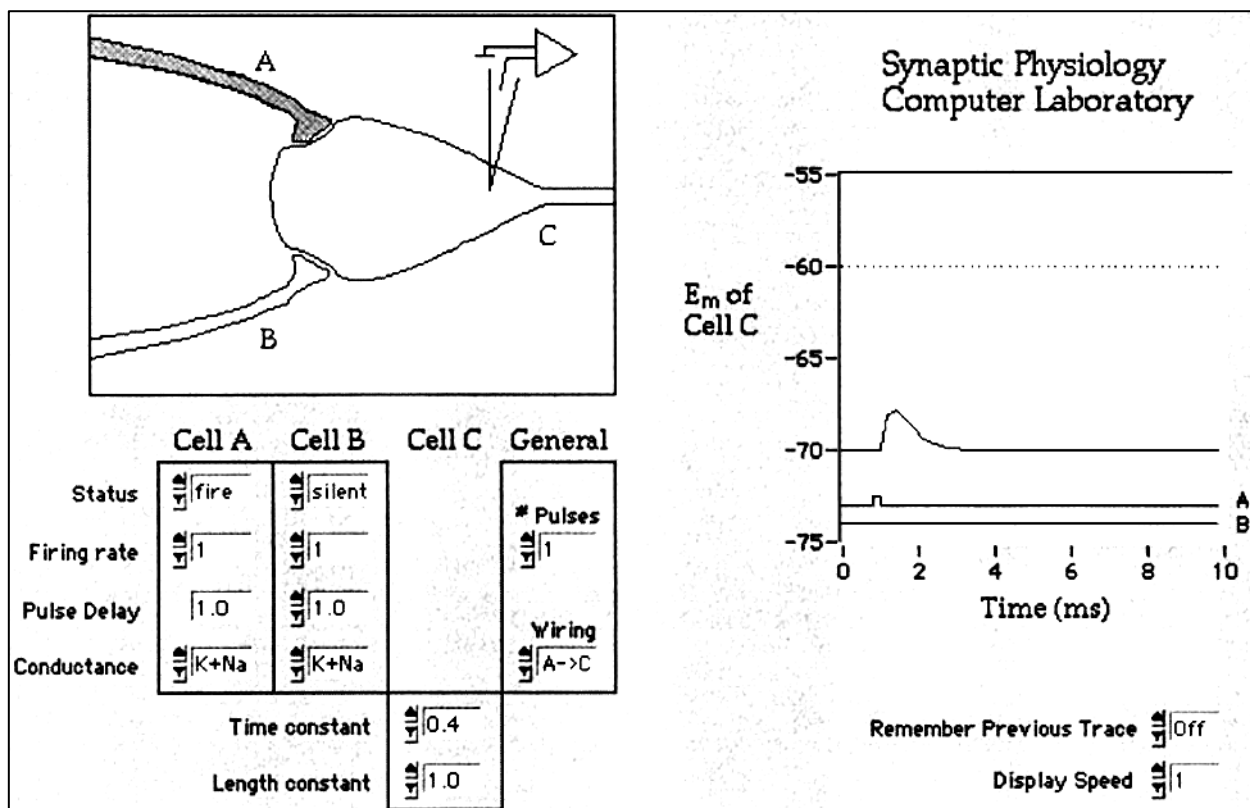


Fig. 6.1

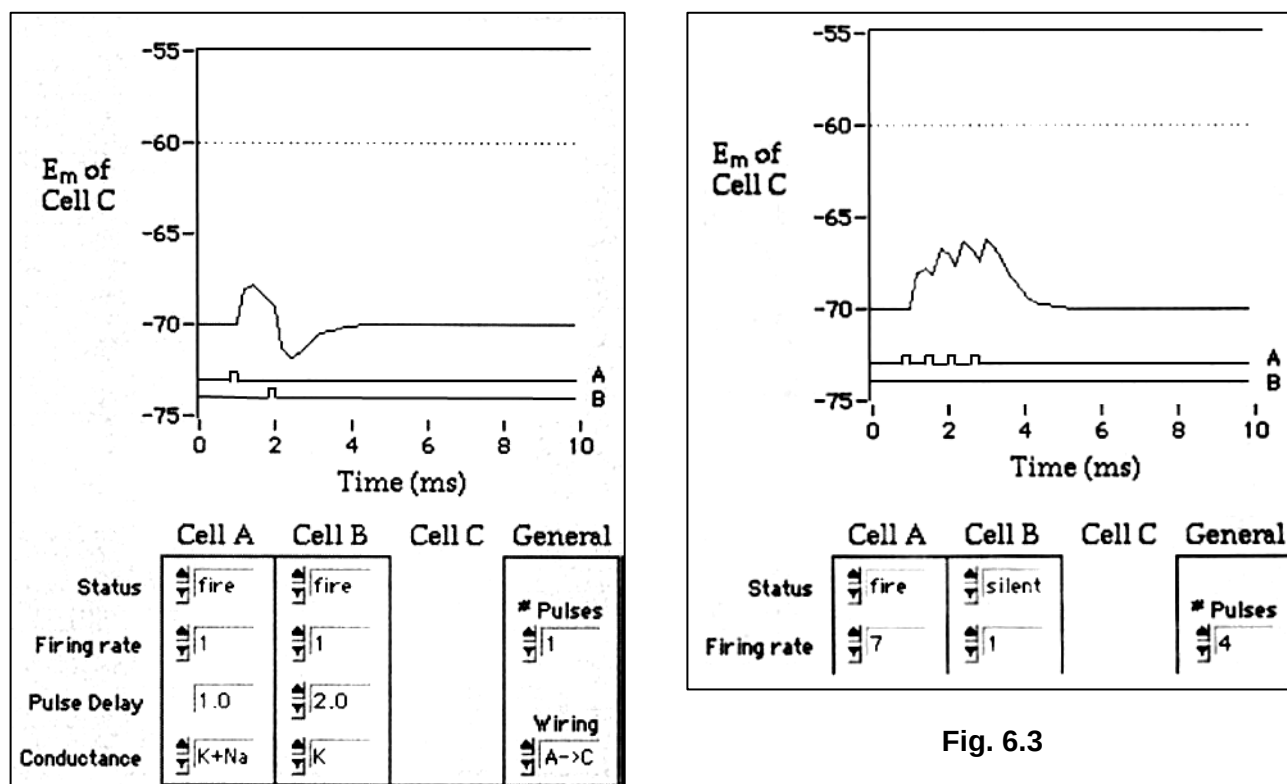


Fig. 6.3

Fig. 6.2

(a) With reference to Fig. 6.1, explain in full, the type of **synaptic** conduction involving Cell A.

.....
.....
.....[2]

(b) With reference to Fig. 6.2, explain in full, the type of **synaptic** conduction involving Cell B.

.....
.....
.....[2]

(c) Explain in full the events occurring in Fig. 6.3.

.....
.....
.....[2]

(d) Explain why demyelinating diseases stop axons from working.

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.....
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.....[4]

[Total: 10]

- 7 Island dwarfism is a phenomenon in which large land dwelling animals which colonize an new island develop into dwarfed versions of the original animal. One animal which has been affected by island dwarfism is the elephant.

Modern elephants belong to two genera (*Elephas* and *Loxodonta*) and are the largest extant (currently living) land animals. African elephants (*L. africana*) can grow to an adult height of 4 metres and weigh up to 9000 kilograms. The Asian Elephant (*E. maximus*) can grow to an adult height of 3 metres and weigh up to 5000 kilograms. These large mammals spend up to 16 hours per day eating plants. An adult African elephant can consume up to 270 kilograms of food each day.

Fossils of various species of extinct dwarf elephants were found on many of the islands in the Mediterranean including Cyprus, Sicily and Crete. These dwarf elephants are believed to have descended from the extinct Straight-Tusked Elephant (*Elephas antiquus*) which inhabited central Europe. Fig. 7 shows the distribution of extinct elephant species in the Mediterranean region and their average masses.

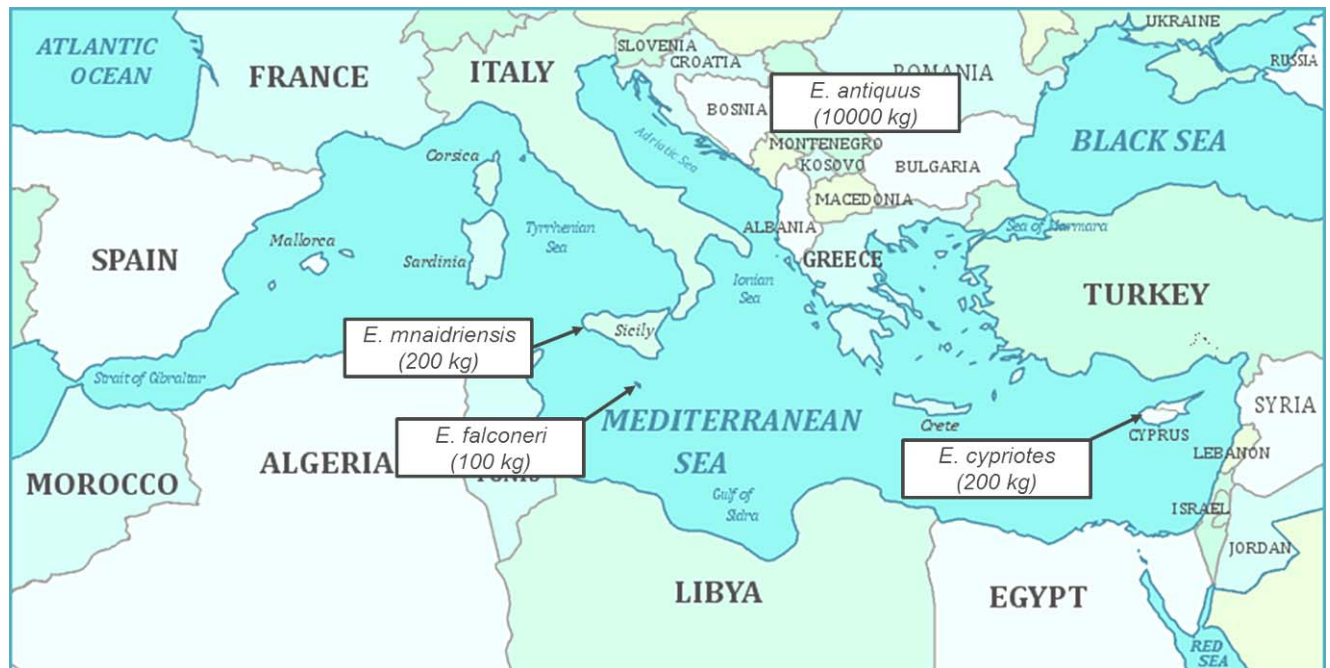


Fig. 7

- (a) (i) State the smallest taxonomic group containing all modern elephants.
.....[1]
- (ii) State the species name of the closest living relative of the dwarf elephants.
.....[1]
- (b) Describe the **biological species concept** and explain whether it is suitable for classifying the dwarf elephants into different species.
.....
.....
.....
.....[2]

- (c) Using neo-Darwinism, explain how founding populations of Straight-Tusked Elephants gave rise to dwarf elephants on the Mediterranean islands.

.....

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.....

.....[4]

- (d) Molecular techniques have been employed to study evolutionary relationships between extinct and extant organisms.

- (i) Explain why DNA analysis is more advantageous than protein analysis as a molecular technique in studying the genus *Elephas*.

.....

.....

.....

.....[2]

- (ii) Briefly describe the neutral theory of molecular evolution and explain how it might be used to study the genus *Elephas*.

.....

.....

.....

.....[2]

[Total: 12]

Section B

Answer **one** question.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8(a)** Describe the pathway taken by cell-membrane bound proteins following synthesis. [6]
- 8(b)** Explain the implication of mutation with respect to phenotype in cystic fibrosis. [7]
- 8(c)** Discuss the significance of the fluid mosaic model of membranes in the light dependent stage of photosynthesis. [7]
- 9(a)** Describe how blood glucose levels are regulated by glucagon via cell signalling pathways. [6]
- 9(b)** Explain the implication of checkpoint dysregulation with respect to cell division in mitosis. [7]
- 9(c)** Discuss the significance of proteins in the control of transcription in eukaryotes. [7]

[Total: 20]

END OF PAPER